## Normal and Hermitian.

**Lemma.** Let $V$ be an inn. Prod space and $T$ is a Hermitian operator.
(If exists) Every eigenvalue of $T$ is real.
Moreover, eigenvectors of $T$ corresponding to distinct eigenvalues are orthogonal.

Proof. Given eigenvalue $c$ of $T$ corresponding to $\alpha$,

$$c\cdot\langle\alpha,\alpha\rangle = \langle T(\alpha),\alpha\rangle = \langle\alpha, T^*(\alpha)\rangle = \langle\alpha, T(\alpha)\rangle = \langle\alpha, c\cdot\alpha\rangle = \bar{c}\cdot\langle\alpha,\alpha\rangle$$

Since $\alpha$ is non-zero, $c=\bar{c}$, or $c$ is a real number.
And, take $\alpha_1,\alpha_2$ be eigenvectors corresponding to $c_1, c_2$, with $c_1\neq c_2$.
Then,

$$c_1\cdot\langle\alpha_1,\alpha_2\rangle = \langle c_1\alpha_1,\alpha_2\rangle = \langle T(\alpha_1),\alpha_2\rangle = \langle\alpha_1, T^*(\alpha_2)\rangle = \langle\alpha_1, T(\alpha_2)\rangle = \langle\alpha_1, c_2\alpha_2\rangle = \bar{c}_2\cdot\langle\alpha_1,\alpha_2\rangle$$

Since $c_2$ must be real, $\bar{c}_2 = c_2$. Now, since $c_1\neq c_2$,

$$(c_1-c_2)\cdot\langle\alpha_1,\alpha_2\rangle = 0 \text{ implies } \langle\alpha_1,\alpha_2\rangle = 0.$$

**Thm.** In finite-dimensional inn. prod space,
Every Hermitian operator has an eigenvector.

Proof. If the given field is $\mathbb{C}$, then the characteristic polynomial must have a root in $\mathbb{C}$ by the Fundamental Theorem of Algebra. So it has an eigenvector.
Now, assume that the given field is the $\mathbb{R}$.
Let $T$ be a Hermitian on $V$ over $\mathbb{R}$, and $\beta$ be an orthonormal basis for $V$.
Let $A=[T]_\beta$. Since $T=T^*=T^t$, $A=A^*=A^t$. Define an inner product on $\mathbb{C}^n$ as:

$$\langle X,Y\rangle \overset{\text{def}}{=} Y^*X$$

Then, a linear operator $L_A:\mathbb{C}^n\to\mathbb{C}^n: X\mapsto AX$ is Hermitian, because

$$\langle L_A(X),Y\rangle = \langle X, L_A^*(Y)\rangle \text{ and } \langle AX,Y\rangle = Y^*AX = \langle X, A^*Y\rangle = \langle X, AY\rangle.$$

By the FTA, $L_A$ has an eigenvalue, and that must be real, by the above Lemma.
Now, $L_A(X)=AX=cX$ for some real number $c$ and a non zero vector $X$ in $\mathbb{C}^n$,
or $\det(cI-A)=0$, $A-cI$ is not invertible, and $A-cI$ has only real-entries,
so $(A-cI)X=0$ has a (non-zero) real solution $X$.

**Lemma.** Let $V$ be a fin. dim inn. Prod space, and $T$ be a linear operator on $V$.
If $W<V$ is $T$-invariant, then $W^\perp$ is $T^*$-invariant.

Proof. Given $\beta\in W^\perp$, and $\alpha\in W$, $\langle T^*(\beta),\alpha\rangle = \langle\beta, T(\alpha)\rangle = 0$, by $T(\alpha)\in W$. ∎

Diagonalizable Criterion of linear operator on inner product space.

Thm. Let $T$ be a linear operator on fin. dim inn. prod space $V$.
If $T$ is Hermitian, then there is an orthonormal basis for $V$ consisting by eigenvectors of $T$.

Proof Let $T$ be a Hermitian. Using induction for $\dim V = n$.
If $n=1$, trivial. And, for $n$, using the facts that:
1). $T$ has an eigenvector $\alpha_1$.
2). $\langle \alpha_1 \rangle \oplus \langle \alpha_1 \rangle^{\perp} = V$.
3). $\langle \alpha_1 \rangle$ is $T$-invariant, so $\langle \alpha_1 \rangle^{\perp}$ is $T^*$-invariant, and $T = T^*$.

Corollary. In $\mathbb{R}$-Inner Product space,
$T$ is symmetric <u>if and only if</u> there is orthonormal basis for $V$ consisting by eigenvectors of $T$.

Lemma. Let $T$ be a normal operator on fin dim inn prod space $V$. Given $\alpha \in V$,
$\alpha$ is eigenvector for $T$ corresponding to $c \in F$ <u>iff</u> $\alpha$ is eigenvector for $T^*$ corresponding to $\bar{c} \in F$.

Proof. If $L$ is normal, then
$\|L(\alpha)\|^2 = \langle L(\alpha), L(\alpha) \rangle = \langle \alpha, L^*L(\alpha) \rangle = \langle \alpha, LL^*(\alpha) \rangle = \langle L^*(\alpha), L^*(\alpha) \rangle = \|L^*(\alpha)\|^2$.
And, any scalar $c \in F$, $T - cI$ is normal, by
$(T-cI)(T-cI)^* = TT^* - \bar{c}T - cT^* + c\bar{c} = T^*T - \bar{c}T - cT^* + c\bar{c} = (T-cI)^*(T-cI)$.

So, $\|((T-cI)(\alpha)\| = \|(T-cI)^*(\alpha)\|$ gives the result.

Thm. Let $T$ be a linear operator on fin.dim V.s $V$, and $\beta$ be an orthonormal basis for $V$.
Suppose that $A = [T]_\beta$ is upper-triangular. Then,
$T$ is normal <u>if and only if</u> $A$ is diagonal.

Proof. Let $T$ is normal and $\beta = \{\alpha_1, \cdots, \alpha_n\}$. Since $A$ is upper triangle,
$T(\alpha_1) = A_{11}\alpha_1$ and so $T^*(\alpha_1) = \bar{A}_{11}\alpha_1 = \sum_{j=1}^{n} (A^*)_{j1} \cdot \alpha_j = \sum \bar{A}_{1j} \cdot \alpha_j$
So, $\bar{A}_{12} = \bar{A}_{13} = \cdots = \bar{A}_{1n} = 0$, Similarly, all non-diagonal entries are zero.
The converse is trivial. ∎
Now, since every linear operator on $\mathbb{C}$-vector space has a basis s.t the representation is triangular
Using Gram-Schmidt process to this, we obtain that every normal in $\mathbb{C}$ is unitary diagonalizable.

Summary.

$T: V \to V$, $V$ fin. dim inn. prod over $F$

$T$ is normal and $F = \mathbb{C}$

$\Updownarrow$

In $\mathbb{C}$, $T$ is Hermitian $\Rightarrow$ $T$ is unitary equivalent to a diagonal matrix

In $\mathbb{R}$, $T$ is symmetric $\Leftrightarrow$ $T$ is orthogonal equivalent to a diagonal matrix

In $\mathbb{C}$, Hermitian $\Rightarrow$ Normal

$\Updownarrow$

Symmetric